Huawei Access Node WHI-ACC-81 Failure SOP: ALARM_9067 - Minor Hardware/Environmental Fault

Node: WHI-ACC-81 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_9067

Description

The failure domain of ASIC memory parity errors, line card crashes, and power supply redundancy loss on the Huawei Access Node WHI-ACC-81 at the Whitefield East POP can be attributed to the complex interplay between the node's hardware components and environmental factors. Specifically, the use of Huawei's ASIC-based architecture, which relies heavily on the Line Card's ability to interface with the Power Supply Unit (PSU) and the System Board, makes it vulnerable to faults in these critical components. The ASIC memory parity errors can occur due to a variety of reasons, including but not limited to, manufacturing defects, overheating, or electrical noise. Similarly, the line card crashes can be triggered by hardware failures, software bugs, or even issues with the node's power supply. Furthermore, the loss of power supply redundancy can have a cascading effect, leading to the failure of other critical components. In order to mitigate these risks, it is essential to perform regular maintenance and inspections, as well as to implement robust redundancy and failover mechanisms. In the event of a failure, prompt diagnosis and resolution are crucial to minimize downtime and ensure continued service availability.

Pre-requisite Checks

- The remote hands team must be dispatched to the Whitefield East POP to perform physical checks and gather data from the node.

- The status of the redundant links must be verified to ensure that they are functioning correctly.

- The node's hardware and software versions must be checked to ensure that they are up-to-date and compatible.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display error-alarm, Expected Output: ALARM_9067: ASIC memory parity error, Description: This command will display the current alarm status and help identify the root cause of the failure.

Command: display power supply, Expected Output: Power supply status: Normal, Description: This command will display the status of the power
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supply unit and help determine if the redundancy loss is related to a
hardware or software issue.

Command: display line card, Expected Output: Line card status: Normal,
Description: This command will display the status of the line card and
help identify if the crash is related to a hardware or software issue.
```

Resolution Steps

Step 1: Step: Replace faulty ASIC or Line Card, Command: replace slot , Description: This step will involve replacing the faulty ASIC or line card with a new one, and then verifying that the node is functioning correctly.

Step 2: Step: Reconfigure power supply redundancy, Command: set power supply redundancy , Description: This step will involve reconfiguring the power supply redundancy level to ensure that the node has a redundant power supply.

Step 3: Step: Update node software, Command: upgrade node software, Description: This step will involve updating the node software to the latest version, and then verifying that the node is functioning correctly.

Post-Check Verification

```
Command: display error-alarm, Expected Output: No alarm status,
Description: This command will verify that the alarm has been cleared
and the node is functioning correctly.

Command: display power supply, Expected Output: Power supply status:
Normal, Description: This command will verify that the power supply is
functioning correctly and redundancy is working as expected.

Command: display line card, Expected Output: Line card status: Normal,
Description: This command will verify that the line card is
functioning correctly and the node is functioning as expected.
```

Roll-back Procedures

- Step: Restore previous configuration, Command: undo set power supply redundancy , Description: This step will involve restoring the previous configuration, including the power supply redundancy level, to ensure that the node is returned to its previous state.

- Step: Undo software update, Command: undo upgrade node software, Description: This step will involve undoing the software update to ensure that the node is returned to its previous state.

- Step: Replace original ASIC or Line Card, Command: replace slot , Description: This step will involve replacing the original ASIC or line card, and then verifying that the node is functioning correctly.